

Entity–Memory Graph Retrieval Improves Evidence Coverage in Long-Conversation Question Answering

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Abstract

Long-term conversational question answering requires a memory system to find relevant evidence across temporally separated dialogue turns. We test whether a conversation-built Entity–Memory graph improves retrieval over a matched dense Memory control when Memory representations, query vectors, answer generation, and evaluation are held fixed. Experiments cover 1,986 question–answer rows from ten LoCoMo conversations. At the primary cutoff *top-k* 25, graph retrieval increases official evidence recall from 79.7468% to 84.4842%. The 4.7374-point difference has paired-question and conversation-cluster 95% bootstrap intervals of 3.6504–5.8425 and 3.5286–6.0124 points. The recall advantage is supported at every preregistered cutoff from 5 to 50, whereas no matched cutoff establishes an overall final-answer F1 difference. Ablations associate the recall gain with semantic scoring, Entity fusion, chronological expansion, and speaker links. A complete cold/warm measurement records construction, indexing, retrieval, provider usage, and storage. Warm retrieval averages 0.004886 seconds per question after all cache-sensitive stages fall to zero new provider requests, compared with 1.5123 seconds in the cold pass. The evidence supports a scoped retrieval claim on the evaluated LoCoMo set. It does not establish improved answer F1, cross-dataset generalization, official-reference parity, or state-of-the-art performance.

Keywords: long-term conversational memory; graph retrieval; evidence recall; retrieval-augmented generation; ablation analysis

1 Introduction

Conversational agents accumulate facts, events, preferences, and relationships over interactions that exceed a model’s usable context. Long-term memory systems must therefore decide what to store, how to organize it, and which parts to retrieve for a new question. The LoCoMo benchmark makes this problem concrete through long, multi-session conversations and questions requiring single-hop, multi-hop, temporal, commonsense, and adversarial reasoning [Maharana et al., 2024]. Long-context and retrieval-augmented generation improve memory-related question answering on LoCoMo, but long-range temporal and causal reasoning remain difficult.

Dense retrieval provides a natural control. It encodes each conversational Memory item, scores that item against the ques-

tion, and sends the highest-ranked items to an answer model. The design is simple and scalable, but it treats Memory items as independent. A graph can represent repeated entities and chronological adjacency, allowing retrieval to use structure that is absent from a flat index. A valid comparison must distinguish that structural contribution from changes in embeddings, context budgets, answer models, or evaluation.

We evaluate an Entity–Memory graph built only from conversation content. Each dialogue item becomes a Memory node, normalized Entities become shared nodes, and chronological links connect neighboring Memories. Entity matching is fused with semantic Memory similarity and chronological expansion at query time. Test questions form retrieval queries only. Questions, answers, evidence labels, categories, predictions, and evaluator output are excluded from graph construction.

This study asks three questions. First, does the graph improve evidence recall over matched dense retrieval when Memory vectors, query vectors, answer generation, and evaluation are fixed? Second, which graph components are associated with any observed difference? Third, does the result persist across retrieval cutoffs and selected input, fusion-weight, and sequence-scale settings? The experiments address these questions through:

- (1) an all-question matched comparison with immutable ordered query vectors and identical answer generation;
- (2) separate measurement of evidence recall and final-answer F1 with paired-question and conversation-cluster uncertainty; and
- (3) component, input, cutoff, parameter, and cold/warm system analyses under frozen protocols.

The result is narrower than a general memory-system claim. Complete graph retrieval improves evidence recall across *top-k* 5, 10, 25, and 50 on the evaluated ten-conversation set, but it does not establish higher overall answer F1.

2 Related Work

2.1 Retrieval foundations and long-context organization

Retrieval-augmented generation combines a parametric generator with retrieved non-parametric evidence [Lewis et al., 2020]. Dense Passage Retrieval showed that a learned dual encoder can retrieve candidate passages for open-domain question answering using dense representations [Karpukhin et al., 2020]. These studies establish the flat dense-retrieval control used in

our matched comparison.

Later systems organize long inputs above the individual-chunk level. RAPTOR recursively embeds, clusters, and summarizes text into a retrieval tree [Sarthi et al., 2024]. MemoRAG uses a long-range model to form global memory and generate clues for a downstream retriever [Qian et al., 2024]. Our method also introduces structure above independent Memory vectors, but it retains dialogue items as the answer evidence rather than replacing them with generated summaries.

2.2 Long-term conversational memory

LoCoMo introduced a human-verified benchmark of long, persona-grounded, multimodal conversations and evaluates question answering, event summarization, and dialogue generation [Maharana et al., 2024]. LongMemEval later framed assistant memory as indexing, retrieval, and reading, and evaluates information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention [Wu et al., 2025]. These benchmarks motivate pipeline-level evaluation rather than attributing all downstream behavior to the language model.

Early agent-memory architectures store experience for later retrieval and reflection. Generative Agents records observations, derives higher-level reflections, and retrieves memories for planning [Park et al., 2023]. MemoryBank adds persistent conversational memory with retrieval, updating, and time-dependent forgetting [Zhong et al., 2024]. MemGPT instead manages limited context through virtual memory and hierarchical tiers [Packer et al., 2023].

Recent systems place more structure in the memory store. Mem0 extracts, consolidates, and retrieves salient conversational information, with a graph variant for relational memory [Chhikara et al., 2025]. A-Mem dynamically indexes and links structured memory notes into an evolving network [Xu et al., 2025]. Our study examines a narrower question than these end-to-end memory architectures. Given a fixed reader and fixed Memory representations, we test whether a conversation-derived graph changes evidence retrieval relative to a flat dense control.

2.3 Graph retrieval

Graph retrieval can expose relations that independent text chunks omit. HippoRAG combines a knowledge graph with Personalized PageRank for long-term knowledge integration and multi-hop retrieval [Gutiérrez et al., 2024]. GraphRAG constructs a graph index and community summaries for global questions over document collections [Edge et al., 2024]. GRAG retrieves textual subgraphs and supplies textual and topological views to a generator [Hu et al., 2025], while LightRAG combines graph indexing with dual-level retrieval over entities and relations [Guo et al., 2025].

Graph structure has also become central to conversational memory. APEX-MEM stores temporally grounded events in a property graph and resolves evolving information at retrieval time [Banerjee et al., 2026]. Mnemis combines a base similarity graph with top-down selection over a semantic hierarchy

[Tang et al., 2026]. GAM separates an event-progression graph from a topic-association network [Wu et al., 2026], whereas TiMem consolidates conversation observations into a temporal hierarchy [Li et al., 2026]. Our graph is smaller in scope. Shared Entity nodes link mentions across dialogue-item Memories, and chronological edges preserve local sequence. The methodological emphasis is a matched retrieval control and the separation of evidence recall from generated-answer F1.

2.4 Dense retrieval controls

DRAGON is a dense retriever trained with diverse data augmentation [Lin et al., 2023] and is used in the original LoCoMo RAG setup. We inspected the pinned upstream reference but did not pass the project’s official reproduction tolerance. A local raw-DRAGON condition is therefore retained only as a diagnostic. It is excluded from controlled and inferential claims. The formal comparison in this paper is internal and matched. Dense Memory retrieval is compared with graph retrieval using identical Memory and query vector artifacts.

3 Method

3.1 Conversation-only graph construction

Figure 1 summarizes the experimental path. For each conversation, the constructor consumes session timestamps, dialogue identifiers, speaker names, dialogue text, and image captions. Each normalized dialogue item becomes a Memory node. A gpt-3.5-turbo extractor at temperature 0.3 identifies Entities from the conversation text. Entity strings are normalized, deduplicated case-insensitively, and connected to every Memory in which they occur. Deterministic speaker links add the dialogue speaker as an Entity. Adjacent Memory nodes are connected in chronological order.

The complete ten-conversation graph contains 5,882 Memory nodes. Graph construction does not consume QA questions, gold answers, evidence annotations, category labels, judge outputs, previous predictions, or question-driven ledgers. This separation prevents test questions from shaping the stored memory structure.

3.2 Retrieval

The question supplies two retrieval-only signals. A question-Entity extractor first produces normalized keys for matching Entity nodes. The question then retrieves its vector from a complete, immutable `text-embedding-3-small` artifact. The artifact is bound to the dataset hash, ordered question digests, model, role, vector dimension, normalization, and artifact SHA-256. Formal retrieval fails on a cache miss and makes no live query-embedding request.

For Memory m , the primary condition combines Entity relevance $s_e(m)$ and signed semantic similarity $s_s(m)$:

$$s(m) = 0.30s_e(m) + 0.70s_s(m). \quad (1)$$

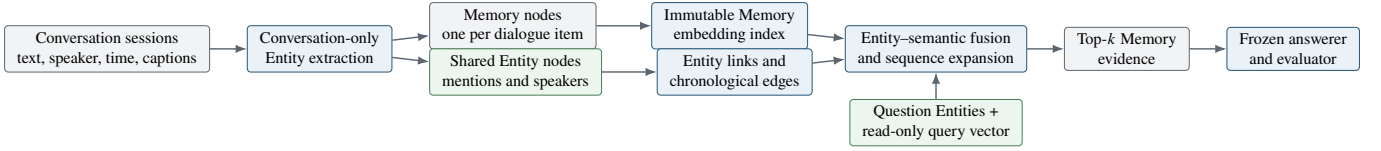


Figure 1: The experimental retrieval path. Graph construction consumes only conversation fields. Questions enter after construction through a question-Entity extractor and a read-only query-vector artifact. The matched dense control uses the same Memory nodes, Memory vectors, query vectors, answerer, and evaluator, but omits Entity fusion and sequence expansion.

Entity candidates must reach a relative-score threshold of 0.5, with at most 20 matches retained per query key. Speaker-only matches are multiplied by 0.25, and Entity contributions are degree-discounted. Chronological neighbors are expanded at secondary scale 0.5. If the gated pool contains fewer than the requested number of Memories, semantic full-pool retrieval fills the remaining positions without discarding gated results. Candidates are ordered by descending fused score and then dialogue id.

The primary context budget is *top-k* 25. Cutoff analyses use *top-k* 5, 10, and 50. The dense control ranks the same Memory representations with the same semantic query vectors but does not use Entity matching or sequence expansion. A dense graph-construction control, B_{embed} , verifies that graph construction alone does not change dense retrieval.

3.3 Answer generation and evaluation

Retrieved Memory text and identifiers enter the frozen LoCoMo-aligned answer interface. The requested answer model is gpt-3.5-turbo, with a system-role prompt, temperature 0, one question per batch, and a 32-token completion limit. The immutable evaluation package retains its category-specific generation, decoding, F1, evidence-recall, rounding, and aggregation behavior.

We report two outcomes. Overall F1 measures final-answer overlap under the frozen evaluator. `recall_acc` is the official evidence-recall definition. The repository-defined Categories 1–4 subset F1 is used only as a diagnostic and is not presented as an official LoCoMo metric.

3.4 Statistical analysis

Every formal condition contains the same 1,986 QA rows from ten conversations. Matched differences are estimated with 10,000 paired-question bootstrap resamples and 10,000 resamples of whole conversations, using seed 20260727. The conversation-cluster estimator reflects uncertainty over the ten conversation units. Component, input, fusion, and sequence families use Holm step-down correction within each outcome and estimator. A difference is treated as supported only when the preregistered evidence gate passes. Failure to reject is not interpreted as equivalence.

Table 1: Primary matched comparison at *top-k* 25. Values are percentages.

Condition	Overall F1	<code>recall_acc</code>
A (dense Memory)	42.0681	79.7468
B_{embed} (dense graph control)	42.0677	79.7468
B (complete graph retrieval)	42.5680	84.4842

4 Experimental Design

The primary comparison holds constant the dataset and row order, Memory text, Memory vectors, immutable ordered query vectors, answer prompt, answer model, token budget, evaluation package, output isolation, and aggregation. The treatment activates Entity matching, graph fusion, and chronological expansion. The B_{embed} control must reproduce the dense control’s ordered context identifiers for all rows before the primary graph contrast is interpreted.

Component conditions remove sequence expansion, Entity-score fusion, or the semantic channel. A speaker ablation removes deterministic speaker links. Input conditions compare time-annotated and raw-text profiles. Sensitivity families vary Entity/Semantic weights and chronological scale. Cutoff analysis uses matched A/B pairs at *top-k* 5, 10, 25, and 50.

Each metric-bearing condition writes to an absent, condition-specific output directory. The record includes the source commit, resolved configuration, model names, artifact identities, query-cache hits and misses, prompt budgets, and graph compliance. The vendored evaluator is verified against its SHA-256 manifest before formal use.

5 Results

5.1 Primary matched comparison

Complete graph retrieval improves `recall_acc` over the matched dense control by 4.7374 percentage points (Table 1). The paired-question 95% interval is 3.6504–5.8425 points, and the conversation-cluster interval is 3.5286–6.0124 points. Both two-sided *p*-values are 0.0002. Overall F1 changes by 0.4998 points, with paired and cluster intervals of −0.5745–1.5514 and −0.1773–1.2711 points. The F1 difference is not supported by either estimator.

The dense graph-construction control and A have zero ordered-context mismatches across all 1,986 rows and identical evidence recall. Their F1 values differ by less than 0.001 per-

centage points, consistent with frozen Category-5 option-order randomness rather than a retrieval difference.

5.2 Cutoff analysis

The evidence-recall advantage persists across all preregistered cutoffs (Table 2). B minus A recall differences range from 3.6159 to 5.5702 points. Paired-question and conversation-cluster intervals are above zero at every cutoff. Both matched F1 intervals include zero at every cutoff. The cutoff analysis therefore supports an evidence-coverage advantage over the tested range, not an answer-quality advantage.

5.3 Components, inputs, and sensitivity

Several graph operations are associated with the retrieval gain (Table 3). Complete B improves recall over the no-sequence condition by 1.6728 points and over the no-Entity-score-fusion condition by 4.7626 points after Holm correction. Neither contrast supports F1. Sequence expansion after Entity gating contributes 0.4364 recall points. The semantic channel relative to Entity-only retrieval contributes 17.4634 recall points and 5.0780 F1 points, with both outcomes supported within the component family. Removing deterministic speaker links lowers recall by 1.4822 points without a supported F1 change.

Input and parameter results bound these component findings. Time annotation has no supported within-B overall effect after correction. When both A and B use raw text, B retains a supported 4.9522-point recall advantage without an F1 advantage. Reducing Entity/Semantic weights from 0.30/0.70 to 0.10/0.90 lowers recall by 2.3062 points after correction. Equal 0.50/0.50 weights are not distinguishable from the primary setting. No overall F1 or recall contrast among sequence scales 0.25, 0.5, and 1.0 survives family correction.

5.4 Cold and warm system cost

The warm replay makes no new provider requests in the three cache-sensitive stages (Table 4). Mean retrieval latency is 309.48 times lower. Cold and warm each cover the same 44 ordered batches, with 1,998 reads from the immutable query artifact, zero misses, and zero live query embeddings. The formal answer trace contains 1,986 requests, 2,744,099 input tokens, and 16,269 output tokens. Because answer generation is independent of cache warmth, the same validated trace is attached to each state rather than rerun.

Graphs occupy 21,689,931 bytes, Memory indexes 29,946,660 bytes, and Entity/question caches 3,411,176 bytes. Monetary cost is not reported because provider pricing was not frozen as an experimental input.

6 Discussion

The experiments separate evidence coverage from answer quality. Entity links, semantic scoring, and chronological expansion retrieve more annotated evidence than the matched dense control, and the difference persists across four context budgets.

The fixed short-answer generator does not convert the additional coverage into a supported overall F1 gain. Evaluations that report only answer scores can therefore obscure retrieval changes, while retrieval improvements should not be described as end-to-end quality gains.

This separation complements LongMemEval’s decomposition of assistant memory into indexing, retrieval, and reading stages [Wu et al., 2025]. Systems such as Mem0, A-Mem, APEX-MEM, Mnemis, GAM, and TiMem jointly change several of these stages [Chhikara et al., 2025, Xu et al., 2025, Banerjee et al., 2026, Tang et al., 2026, Wu et al., 2026, Li et al., 2026]. Their published scores therefore answer broader system-level questions than our matched contrast. Differences in readers, graph construction, prompts, context budgets, and metrics preclude a numerical ranking against our result. Our evidence instead identifies what changes when the stored Memory text, Memory vectors, query vectors, answerer, and evaluator remain fixed.

The component results suggest that the complete difference is distributed across the retrieval path. The semantic channel is the largest isolated contributor. Entity fusion, sequence expansion, and speaker links add smaller supported recall contributions. The raw-text comparison indicates that the B-over-A recall advantage does not depend on the tested time annotation. Sensitivity results also discourage presenting the primary parameters as universally optimal. One lower-Entity-weight condition reduces recall, an equal-weight condition is not distinguishable, and no tested sequence scale survives family correction.

The cold/warm result exposes an operational tradeoff. Conversation Entity extraction and Memory embedding are one-time costs, while uncached question-Entity extraction dominates cold retrieval latency. Once the conversation artifacts and question-Entity cache are present, the graph path requires no new provider request before answer generation and retrieval takes milliseconds per question. This result is relevant for repeated evaluation and repeated queries, but it does not imply the same latency on another machine, provider, or workload.

6.1 Limitations and future work

First, formal evidence comes from ten conversations in one LoCoMo release. Conversation-cluster bootstrap reflects uncertainty over these ten units but does not establish cross-dataset generalization. Replication on additional conversation-memory benchmarks is needed.

Second, no matched cutoff supports an overall final-answer F1 difference. The contribution concerns retrieval coverage, not a general improvement in generated answers. Alternative readers and evidence selectors require separately controlled experiments.

Third, the official DRAGON reproduction gate did not pass. The local raw DRAGON experiment changes semantic and query artifacts and remains diagnostic. A future official-reference comparison requires parity with the pinned upstream implementation.

Fourth, the requested answer-model name is recorded, but

Table 2: Matched cutoff analysis. Values are percentages; differences are B minus A in percentage points. Every paired-question and conversation-cluster recall interval is above zero. Every corresponding F1 interval includes zero.

<i>top-k</i>	A F1	B F1	Δ F1	A recall	B recall	Δ recall	Paired recall CI	Cluster recall CI
5	39.9053	39.8264	-0.0790	59.3584	64.1667	+4.8083	[3.3450, 6.2808]	[2.8809, 6.6134]
10	41.7607	42.1412	+0.3805	68.9145	74.4847	+5.5702	[4.2928, 6.8624]	[4.2365, 6.7522]
25	42.0681	42.5680	+0.4998	79.7468	84.4842	+4.7374	[3.6504, 5.8425]	[3.5286, 6.0124]
50	42.0782	41.7832	-0.2950	86.7148	90.3306	+3.6159	[2.7754, 4.4735]	[2.4603, 4.6426]

Table 3: Component and input contrasts at *top-k* 25. Differences are treatment minus control in percentage points. “Supported” refers to the preregistered family-corrected inference unless noted otherwise.

Contrast	Δ F1	Δ recall_acc	Conclusion
B minus B without sequence expansion	+0.5137	+1.6728	Recall supported; F1 not supported
B minus B without Entity-score fusion	+0.8014	+4.7626	Recall supported; F1 not supported
Sequence expansion after Entity gating	-0.0609	+0.4364	Recall supported; F1 not supported
B minus Entity-only retrieval	+5.0780	+17.4634	Both outcomes supported
Raw-text B minus annotated-input B	+0.2407	-0.2472	Neither outcome supported
Raw-text B minus raw-text A	+0.3508	+4.9522	Recall supported; F1 not supported
B minus B without speaker links	+0.2415	+1.4822	Recall supported; F1 not supported

the provider did not persist a separate actual-model identity in the formal snapshots. Category 5 also retains the frozen upstream unseeded option-order behavior. Future runs should preserve provider-returned model identities when available and quantify Category-5 generation variance separately.

Finally, system timing was measured on one machine and one provider endpoint. The warm condition assumes complete reusable graph, index, and question-Entity artifacts. Replication across hardware and providers is needed before deployment-level latency claims.

7 Reproducibility and Responsible Use

Every formal condition records its source commit, resolved configuration, dataset and artifact SHA-256 values, isolated output path, query-cache usage, provider-token telemetry, graph-input audit, and prompt budget. Query vectors are complete and read-only during formal retrieval. The evaluation package is vendored and hash-verified. Statistical reports and the final cost report have independent byte-identical reproductions.

The graph uses conversation data, including speaker names and image captions, which can contain personal information in real deployments. Systems applying this method outside the benchmark should establish consent, retention, deletion, and access-control policies. The present work evaluates benchmark retrieval and does not study privacy attacks or sensitive-memory deletion.

8 Conclusion

Long-conversation question answering requires retrieval that preserves relevant evidence without using test questions to

build memory. Graph retrieval may represent links that a flat index omits, but a fair test must hold the dense representations, query vectors, answer generation, and evaluation fixed. Our matched design addresses this evidence gap.

The Entity-Memory graph improves official evidence recall over dense Memory retrieval across *top-k* 5–50 on the evaluated LoCoMo set. Component analyses associate the recall difference with semantic scoring, Entity fusion, chronological expansion, and speaker links. The same experiments do not establish higher overall final-answer F1. Finally, the cold/warm measurement shows that reusable graph, index, and question-Entity artifacts remove new provider requests from the retrieval path before answer generation.

This work is best viewed as a scoped retrieval study. The ten-conversation sample, failed official DRAGON reproduction gate, fixed reader, and single-environment latency measurement bound the claim. Separating evidence coverage from answer quality provides a transferable evaluation principle for future long-term conversational memory systems.

Data and Code Availability

LoCoMo data are available from the benchmark authors [Maharana et al., 2024]. Code, experiment manifests, validation records, and manuscript evidence snapshots for this study are available at https://github.com/Sun668/graph_memory. The exact manuscript snapshot corresponds to the repository history recorded with the arXiv source bundle.

Table 4: Primary-B cold and warm system measurement. Stage wall times overlap and must not be summed as end-to-end latency.

Stage	Cold req.	Cold tokens (input/output)	Cold timing	Warm req.	Warm timing
Conversation Entity extraction	5,873	3,731,393 / 425,637	9,792.76 s summed call time	0	0.00 s
Memory embedding	591	214,229 / 0	443.38 s	0	1.29 s cache load
Question Entity extraction	1,974	1,214,259 / 104,272	2,970.83 s	0	0.00 s
Retrieval (1,986 QA)	0	0 / 0	mean 1.5123 s; p95 2.6132 s	0	mean 0.004886 s; p95 0.006270 s

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A Resolved Primary Configuration

Table 5: Resolved settings for the primary B condition.

Dimension	Resolved value
Dataset	LoCoMo ten-conversation release; 1,986 QA rows
Graph inputs	Session time, dialogue id, speaker, text, image caption
Excluded graph inputs	Questions, answers, evidence, categories, judges, predictions
Memory nodes	5,882
Conversation Entity extractor	gpt-3.5-turbo; temperature 0.3
Memory embedding model	text-embedding-3-small
Answer model request	gpt-3.5-turbo; temperature 0; 32 output tokens
Primary context budget	<i>top-k</i> 25
Entity / semantic weights	0.30 / 0.70
Entity relative-score threshold	0.5
Maximum matches per Entity key	20
Speaker-only dampening	0.25
Degree discount	Enabled
Sequence expansion scale	0.5
Query embeddings	Complete read-only artifact; L2-normalized float32
Bootstrap	10,000 paired QA and 10,000 conversation-cluster resamples
Bootstrap seed	20260727

B Validity and Reproducibility Gates

Table 6: Checks applied before promotion of formal evidence.

Gate	Requirement	Outcome
Graph construction	Conversation-only inputs; no QA-derived construction fields	Pass
Evaluator integrity	All 16 vendored files match the frozen SHA-256 manifest	Pass
Output isolation	One absent or empty directory per resolved formal condition	Pass
Query vectors	Complete ordered coverage; zero miss; zero live query embedding	Pass
Dense control	A and B_{embed} ordered contexts equal for all 1,986 rows	Pass
Prompt budget	Non-data scaffold no longer than 5,000 characters	Pass (2,410)
Statistical reproduction	Independent report build is byte-identical	Pass
Cost reproduction	Independent cost report build is byte-identical	Pass
Cache preservation	Quarantined artifacts retain frozen identities	Pass

C Claim Boundary

The strongest supported statement is that complete conversation-built Entity–Memory retrieval improves official evidence recall over matched dense Memory retrieval across *top-k* 5, 10, 25, and 50 on the evaluated ten-conversation LoCoMo set. The evidence does not support an overall final-answer F1 advantage, cross-dataset generalization, an official DRAGON comparison, state-of-the-art status, or monetary cost. These exclusions apply to the Abstract, Results, Discussion, and Conclusion.